

CS 430

SLCM 0-5 Population Project — Part 1

Written Report

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What did you get to know?

What surprised us the most was how many families in the dataset are dealing with housing problems. Many households have past-due rent, and some are currently being evicted or are unhoused. We were also surprised by the household information. The average household size was about 3 people, and families had an average of about 2 children. The household member data also showed that many of the additional family members were young children.

Before doing this project, I did not realize how connected housing problems and family needs are for SLCM's 0–5 population. After looking at the data, I believe many families with young children are dealing with more than one challenge at the same time. Housing stability, income, employment, and family size can all affect the support a family may need. I also learned that

some families come back for help more than once. We found 27 people with multiple profile tickets, and some of their housing or eviction situations changed over time.

One question I still cannot answer with this data alone is **why some families are able to improve their housing situation while others continue to struggle**. We would need more information about the services families received and what happened after receiving help to answer that question. Another question we still can't answer with this data alone is how many families who first connected with SLCM ever advanced into a longer-term program or into the kind of sustained, policy-level support their Rebuild stage describes. Nothing in the ticket data tracks a case's status once the initial intake is logged, and PID's reliance on name-matching means we can't even fully trust our own count of who came back. Answering this would require SLCM to start tracking a case's current program stage over time, along with a more reliable way to link a person's visits together, exactly the two gaps we identified in this analysis.

The Five Questions

1. Housing crisis severity: what share of tickets show active eviction risk (currently being evicted, or unhoused) versus stable housing? Does this differ by zip code or by household size?

Overall crisis rate: 28.2% of the 426 profile-ticket families show active housing crisis (currently being evicted or unhoused), while 71.8% show stable housing at the time of intake.

By household size: crisis rate generally rises with household size, from 19.2% for households with no additional members to 36.9% for households with 4 additional members. Sizes above 4

(5 through 8 additional members) show more erratic rates (15% to 60%), but each of those groups has only 5-20 families, too few to treat as a reliable pattern.

By zip code: 40214 and 40215 are by far the largest zip codes in the data (102 and 89 families respectively, nearly half of all cases combined) and both show relatively lower crisis rates, 20.6% and 13.5%. Smaller zip codes show more extreme rates in both directions (several at or near 100%, several at 0%), but most of these represent very small samples, often fewer than 10 families, so we treat those as unreliable rather than meaningful geographic differences.

Data quality note: the raw zip code field originally contained ZIP+4 formatting inconsistencies and at least one apparent typo. These were cleaned by standardizing to 5-digit zip codes before this analysis.

2. Employment vs. income mismatch: employment status alone doesn't predict financial hardship in this data. Show, with a plot, how income amount and source of income vary between employed and unemployed households.

Employed households report a higher median income (\$2,000/month) than unemployed households (\$0/month). However, employment status alone does not predict financial stability: 30.9% of employed households still rely on food benefits (TANF/SNAP) as an income source, compared to 50.7% of unemployed households. Unemployed households are far more likely to report no income at all, 32.4% versus just 1.1% for employed households. A few things are worth keeping in mind about this comparison. We excluded 29 profiles (7%) with an "Other" employment status, since their income sources were mostly Social Security, food benefits, or no

income at all, suggesting this group represents people outside the employed/unemployed framework entirely. We also used median instead of mean income, since a handful of very high, round-number values among employed households, \$75,000, \$30,000, and \$29,000, appear to be annual salaries entered into what should have been a monthly income field. These skew the mean upward, while the median isn't affected by them. Finally, three unemployed households had no income source listed and were left out of the income-source percentages, so those percentages reflect 219 households rather than the full 222.

3. Household composition: what does the typical household look like in this data (household size, number of children, ages)? Use `household_members_long_v2.csv` to describe the additional family members, not just the primary contact.

The typical household in this dataset has about 3 people. The average household size is 3.33, and the median is 3 people. Households range from 1 to 9 people. Families have an average of 2.16 children, with a median of 2 children. The additional household members are mostly young, with an average age of 9.37 and a median age of 6. Most of the identified household members are sons and daughters. Other relationships include spouses, husbands, mothers, grandsons, boyfriends, wives, and fiancés. One limitation is that 361 relationship values are missing, so we cannot fully describe every household.

4. Repeat contact: using PID, how many people appear on more than one ticket? For people with 2+ tickets, does anything change between their first and most recent ticket (housing status, eviction status)?

There were 27 people who appeared on two or more profile tickets. When we compared their first ticket to their most recent ticket, 5 people had a change in housing status and 8 people had a change in eviction status. Most repeat-contact households did not have a recorded change in these two fields. This shows that some families' housing situations changed between contacts, but most stayed the same based on the available data.

Caveat:

One limitation is that some statuses are recorded as “Unknown,” so a change in the data does not always mean the family's actual situation became better or worse. It may also mean that more information was available at a later contact.

5. Data quality: find and document at least two data quality issues in this dataset.

Explain how you would handle each one and why.

Issue 1: Unknown eviction status. There are 63 records where the eviction status is listed as “Unknown.” I would keep these records, but I would not treat them as either being evicted or not being evicted. This prevents us from making assumptions about information we do not know.

Issue 2: Missing ZIP codes. There are 3 records with missing ZIP codes. I would keep these records for the overall analysis, but leave them out when comparing results by ZIP code. This allows us to still use the other information from these records.

Another data quality issue: There are 31 duplicate PIDs. These should not automatically be deleted because the same person can have more than one ticket. We should use the PID to identify repeat contacts and compare their first and most recent tickets.